PROJECT DOCUMENTATION: Telecom Customer Churn Prediction System

Project Title: Telecom Customer Churn Prediction System

Duration: 3 Weeks (November 10-29, 2025)

Status: Production-Ready with Docker Deployment

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GitHub Repository:
`https://github.com/noushad999/telecom-churn-prediction.git`

SECTION 1: PROJECT OVERVIEW & OBJECTIVES

1.1 Business Problem

Challenge:

- Telecom companies lose billions annually due to customer churn.
- **Current Approach:** Reactive (trying to fix it after the customer leaves).
- **Need:** Proactive identification of at-risk customers (30-90 days in advance).

Market Opportunity (Example: Grameenphone):

- **Subscribers:** 15M+
- **Average Churn Cost:** $100 - $500 per customer.
- **Total Annual Risk:** $1.5B - $7.5B.

Solution: An AI system to predict churn, automate retention strategies, and provide real-time decision support.

1.2 Project Objectives

Primary Goals:

- Build ML model achieving **>70% accuracy**.
- Create a production-ready **REST API**.

- Deploy using **Docker** containerization.
- Generate actionable business intelligence.
- Quantify business impact (**$2M+ savings potential**).

Technical Goals:

- Process **10,000+** customer records.
- Handle class imbalance (**30% churn** vs 70% non-churn).
- **<100ms** prediction latency.

Business Goals:

- Identify **1,700+** high-risk customers.
- Enable **50% retention** (= $2.1M savings).
- Achieve **738% ROI** on retention campaigns.

1.3 Key Metrics & Success Criteria

Category	Metric	Result	Target
Model Performance	Accuracy	**74.96%** ✅	>70%
	Precision	**75.08%** ✅	Minimize False Positives
	Recall	**74.96%** ✅	Catch Churners
	F1-Score	**0.7496** ✅	Balanced Metric
	AUC-ROC	**0.82** ✅	Good Discrimination
System Performance	API Response	**<100ms** ✅	Fast
	Model Size	**752KB** ✅	Lightweight
Business Impact	Revenue at Risk	**$4.2M**	Identified
	Potential Savings	**$2.1M**	@ 50% Retention
	ROI	**738%**	Campaign Return

SECTION 2: DATA ANALYSIS & PREPROCESSING

2.1 Dataset Overview

- **Source:** Telco Customer Churn (Kaggle)
- **Records:** 10,000 customer records (Synthetic historical data)
- **Features:** 19 total (11 Categorical, 3 Numerical)
- **Target Variable:** Churn (Yes/No)
- **Initial Distribution:** 7,139 No (71.39%) vs 2,861 Yes (28.61%) → *Imbalanced*

2.2 Exploratory Data Analysis (EDA) - Key Findings

1. **Contract Type Impact:**
    - Month-to-month: **45% churn rate** (High Risk)
    - Two-year: **3% churn rate** (Low Risk)
2. **Tenure Impact:**
    - <12 months: **50% churn rate**
    - 24 months: **10% churn rate**
3. **Tech Support:**
    - Without tech support: **42% churn**
    - With tech support: **15% churn** (Impact: +27% retention)
4. **Internet Service:**
    - Fiber optic users have a **42% churn rate** (Indicates service quality issues).

2.3 Data Preprocessing Pipeline

Step 1: Data Cleaning & Encoding

- Removed `customerID`.
- Handled missing values.
- **Label Encoding:** Converted categorical text to numbers (e.g., Female: 0, Male: 1).

Step 2: Feature Scaling

- Used `StandardScaler`.
- **Why:** Neural networks learn better with normalized inputs (Mean: 0, Std Dev: 1).

Step 3: Class Balancing (ADASYN)

- **Problem:** Only 28% churners caused bias toward "No Churn".
- **Solution:** Used **ADASYN** (Adaptive Synthetic Sampling).
- **Result:**
    - Before: 7,139 No / 2,861 Yes
    - **After:** 7,139 No / **6,837 Yes** (Ratio ~1:1)

Step 4: Train-Test Split

- **Ratio:** 80% Training / 20% Testing.
- **Stratified:** Maintained class distribution in both sets.

SECTION 3: MACHINE LEARNING MODEL ARCHITECTURE

3.1 Model Selection

After comparing Logistic Regression (68%), Random Forest (71%), and XGBoost (72%), **Deep Neural Network (DNN)** was selected for the best balance of accuracy (74.96%) and inference speed (<50ms).

3.2 Neural Network Architecture

- **Input Layer:** 19 features (Fully Connected)
- **Hidden Layer 1:** 256 neurons + ReLU + Batch Norm + 30% Dropout
- **Hidden Layer 2:** 128 neurons + ReLU + Batch Norm + 30% Dropout
- **Hidden Layer 3:** 64 neurons + ReLU + Batch Norm + 30% Dropout
- **Output Layer:** 2 neurons (Softmax Activation) → `[P(No Churn), P(Churn)]`
- **Total Parameters:** 47,298

3.3 Training Configuration

- **Optimizer:** Adam (Learning Rate: 0.001)
- **Loss Function:** CrossEntropyLoss
- **Batch Size:** 64
- **Epochs:** 50 (with Early Stopping at patience 10)

3.4 Model Evaluation (Test Set)

Confusion Matrix:

- **True Negatives:** 1037 (Predicted Stay, Actually Stayed)
- **True Positives:** 1059 (Predicted Churn, Actually Churned)
- **False Positives:** 391
- **False Negatives:** 309

Interpretation:

- **Precision (73%):** When we say they will churn, we are right 73% of the time. (Saves budget).
- **Recall (77%):** We catch 77% of all actual churners. (Minimizes missed opportunities).

SECTION 4: REST API DEVELOPMENT

4.1 API Architecture

- **Framework:** Flask (Python)
- **Endpoints:**
    1. `GET /` : API Documentation
    2. `GET /health` : System health check
    3. `POST /predict` : Single customer prediction
    4. `POST /predict_batch` : Batch processing

4.2 Prediction Implementation (Code Logic)

1. Receive JSON data.
2. Convert to DataFrame & Encode features.
3. Scale features using saved `StandardScaler`.
4. PyTorch Inference (Forward pass).
5. Apply `Softmax` to get probability.
6. **Logic:**
    - Prob < 0.3: **Low Risk**
    - Prob < 0.7: **Medium Risk**
    - Prob > 0.7: **High Risk**
7. Return JSON response with **Risk Level** and **Actionable Recommendation**.

4.3 Performance

- **Average Latency:** 95ms
- **Throughput:** ~600 predictions/minute (Single process)
- **Memory Usage:** <150MB overhead

SECTION 5: DOCKER DEPLOYMENT

5.1 Docker Strategy

Why Docker? Solves the "It works on my machine" problem by bundling Python 3.10, PyTorch, and all dependencies into a single image.

5.2 Dockerfile Breakdown

Dockerfile

```
# Base Image
FROM python:3.10-slim

# Install System Dependencies
RUN apt-get update && apt-get install -y gcc g++

# Install Python Dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy App Code
COPY . .

# Expose Port & Command
EXPOSE 5001
CMD curl -f http://localhost:5001/health || exit 1
```

5.3 Deployment Steps

1. **Build:** `docker-compose build --no-cache`
2. **Run:** `docker-compose up -d`
3. **Verify:** `docker ps` (Check if status is "Healthy")
4. **Test:** `curl http://localhost:5001/health`

SECTION 6: BUSINESS INTELLIGENCE & ROI

6.1 Risk Segmentation (10,000 Customers)

- **Low Risk (48.7%):** Monitor monthly.
- **Medium Risk (33.9%):** Proactive weekly engagement.
- **High Risk (17.4%): URGENT** intervention within 48 hours.

6.2 Financial Impact Analysis

- **Total Revenue at Risk: $4.2M / year**
- **Campaign Budget:** $430,750 (Intervention costs)

ROI Scenarios:

1. **Conservative (30% Retention):** Net Benefit $390K (**90% ROI**)
2. **Expected (50% Retention):** Net Benefit $937K (**218% ROI**)
3. **Optimistic (70% Retention):** Net Benefit $1.48M (**345% ROI**)

SECTION 7: TECHNOLOGY STACK

Category	Tool	Purpose
ML Framework	**PyTorch**	Neural Network training & inference
Web Framework	**Flask**	REST API creation
Containerization	**Docker**	Reproducible deployment
Data Processing	**Pandas**	Data manipulation & analysis
Version Control	**Git/GitHub**	Source code management
Sampling	**ADASYN**	Handling Class Imbalance

SECTION 8: INTERVIEW QUESTION BANK (Q&A)

PART A: PROJECT OVERVIEW (General)

Q1: Tell me about your telecom churn prediction project.

> **Answer:** "I built a production-ready machine learning system to predict telecom customer churn with **74.96% accuracy**. The system identifies at-risk customers 30-90 days in advance.
>
> **Key highlights:**
>
> - Trained a deep neural network on 10,000+ customer records.
> - Handled class imbalance using ADASYN.
> - Created a REST API with <100ms latency.
> - Deployed using Docker.
> - Identified **$4.2M** annual revenue at risk and proposed strategies with a potential **738% ROI**."

Q2: What was the business problem?

> **Answer:** "Telecom companies operate on a reactive model—fixing issues only after a customer leaves. For a company like Grameenphone with 15M users, even 1% churn costs millions. My solution shifts this to a **proactive** model, identifying high-risk customers early so the marketing team can offer retention incentives *before* the churn happens."

PART B: TECHNICAL DEEP-DIVE

Q5: Tell me about your Neural Network Architecture. Why did you choose it?

> **Answer:** "I used a 3-layer Deep Neural Network (DNN).
>
> - **Input:** 19 features.
> - **Hidden Layers:** 256 → 128 → 64 neurons (Pyramid structure to distill features).
> - **Components:** Each layer uses **ReLU** (for non-linearity), **Batch Normalization** (for faster training), and **30% Dropout** (to prevent overfitting).
> - **Output:** Softmax for probability distribution.

I chose DNN over Random Forest because it offered higher accuracy (74.96% vs 71%) and, crucially, faster inference speed (<50ms) which is vital for real-time API calls."

Q6: How did you handle Class Imbalance?

Answer: "The dataset was imbalanced with only 28% churners. If I trained on raw data, the model would be biased toward 'No Churn'. I used **ADASYN (Adaptive Synthetic Sampling)**. Unlike simple oversampling (which duplicates data), ADASYN generates synthetic data points specifically in 'hard-to-learn' regions. This brought my class ratio to ~1:1 and improved Recall from near 0% to **77%**."

Q7: Why did you use F1-Score and AUC-ROC instead of just Accuracy?

Answer: "Accuracy can be misleading in imbalanced datasets. A model could predict 'No Churn' for everyone and still get 71% accuracy, but it would be useless for business.

- **Recall (77%):** Ensures we catch the actual churners.
- **Precision (73%):** Ensures we don't waste budget on loyal customers.
- **AUC-ROC (0.82):** Proves the model is excellent at distinguishing between churners and non-churners."

PART C: DEPLOYMENT & SYSTEMS

Q9: Why did you use Docker?

Answer: "To solve the 'it works on my machine' issue. My laptop runs Windows/Python 3.10, but production servers might run Linux/Python 3.8. Docker bundles the OS, Python version, PyTorch, and all libraries into a single image. If it runs in Docker, it runs anywhere (AWS, Azure, or On-Premise) without dependency conflicts."

Q12: What is the API latency? How did you optimize it?

Answer: "The target was **<100ms**. My average latency is **95ms**. **Optimization Strategy:**

1. **CPU Inference:** For single predictions, CPU is faster than GPU (avoids data transfer overhead).
2. **Model Size:** Kept the model lightweight (47K parameters, 752KB).
3. **Efficient Preprocessing:** Optimized the encoding and scaling pipeline."

PART D: SOFT SKILLS & VALUE

Q14: Why should we hire you for this role?

Answer: "Because I don't just build models; I build **solutions**.

1. **Business Mindset:** I calculated the $4.2M risk and the ROI of the solution. I speak the language of business, not just code.
2. **Full-Stack ML:** I didn't stop at a Jupyter Notebook. I built an API, containerized it with Docker, and created documentation. I can deploy what I build.
3. **Problem Solver:** I independently tackled class imbalance with ADASYN and optimized latency for production.

I am ready to bring this end-to-end engineering mindset to your team immediately."

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